SPX Local Volatility and Chronological Delta Hedging

Calibration, numerical verification and the 2013–2023 retrospective study

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1 Abstract

I construct daily SPX volatility surfaces and evaluate chronological, fixed-contract option hedges from 2013 to 2023. Andreasen–Huge (AH) calibration supplies a normalized European price surface and a recovered local-variance function. A backward finite-difference solver calculates prices and spot Greeks. Black, smile-based and past-only empirical hedges are compared on observed contract paths under a common self-financing ledger.

The completed retrospective study selected 44,500 entries on 2,768 reference dates and recorded 3,879,048 completed strategy/schedule/scenario outcomes. For five-session holdings, daily rebalancing and midpoint execution, AH Gain was −25.58% on its candidate–Black pairs and −30.16% on the intersection shared by all methods. The latter represents 30.16% more squared funded error than Black, with a 95% interval for Gain of [−55.50%, −7.57%] at mean bootstrap block 20 sessions. AH point estimates are negative across all fifteen declared schedule/execution combinations. A close daily fit to European prices did not translate into a stable historical hedge advantage under these conventions.

Selected numerical panels support price and delta stability within their recorded scope; gamma remains sensitive to spatial position and short-end blending. Thirty dates have unavailable carry before AH optimization. The 31 bad expiry records and 72 scalar ledger replays of selected tail paths are diagnosed from saved cache without changing the experiment. The results are research hedge-error comparisons with provisional contract identity, assumed rates and synthetic execution, rather than executable investment returns.

2 Research question and experiment design

The question is whether a surface that matches current option prices produces better subsequent hedges than Black delta. It separates three tasks: fitting a cross-section of European prices, solving the resulting pricing model accurately, and predicting how market prices change over later sessions. Those tasks require different evidence. Calibration residuals assess fit, analytical controls and refinements assess numerical behavior, and chronological funded hedge errors assess the historical comparison.

For each entry, I retain the option's root, expiry, strike and payoff kind. Rebalancing uses the next current-date model and quote; it does not roll to a new nearest-at-the-money contract. The declared holding/rebalance schedules are 1/1, 5/1, 5/5, 10/1 and 10/5 reference sessions. The source data, selection, model specification and outcomes were frozen before aggregate interpretation. No historical outcome was revised for the publication.

The entire declared dataset is retrospective. September and October 2023 were inspected during development, and additional dates were used for numerical follow-up. Past-only empirical estimation respects information chronology at each prediction, but does not turn the broader study into an untouched holdout.

3 Data, time and financial conventions

The input comprises 132 monthly files with 14,750,241 raw rows. Call and put sides are separated during preparation. These raw rows are neither the selected entry count nor independent hedge comparisons. A cash-market reference calendar contains 2,768 sessions; 24 lack observations. Twenty-one observed early-close dates are excluded under the provisional clock policy.

Preparation checks the Unix and readable timestamps, stated quote date, index mark, contract fields, duplicate keys and bid–ask ordering. Invalid or locked observations remain classified. Entry selection requires positive, two-sided research quotes. Eligible zero-bid endpoint bounds are

handled separately; missing prices are never replaced by a model price or carried forward to an adjacent date.

The files do not establish certified SPX/SPXW roots and official settlement identity for each observation. The research root is `UNKNOWN`, and the assumed fixing is 16:00 New York PM fixing. Maturity uses elapsed seconds to that fixing on ACT/365F, including the recorded timezone and daylight-saving transitions. This is a research convention, not a claim about every contract's official settlement specification. Paths are closed before expiry; settlement payoffs are not substituted for observed endpoint marks.

Convention	Frozen treatment
Calibration domain	Eligible research quotes within 4–60 calendar days and $\lvert\log(K/S)\rvert \le 0.1$
Entry basket	Nearest 21, 35, 45 day expiries; log spot-moneyness -0.02, 0, $+0.02$; calls and puts
Entry eligibility	14–45 calendar days, current-date information, deterministic tie breaking
Contract key	Root, expiry, strike, payoff kind
Pricing discount	Assumed constant 5% rate
Pricing forward	Expiry-specific put–call parity fit
Cash funding	Constant 5% lending and borrowing
Hedge instrument	Fractional synthetic index at observed index mark
Hedge-account dividend cash	Zero
Execution cases	Midpoint; option bid–ask; option bid–ask plus 1 bp hedge-notional fee

Pricing yield inferred from the forward curve is distinct from cash dividends credited to the hedge account. This distinction is preserved in the code and in interpretation. The synthetic hedge has no verified executable index spread, futures basis or discrete contract rounding.

4 Notation and the local-volatility model

Fix a valuation date as time zero. Let S_t be the index, K the strike, t elapsed model time and T expiry, in years. Prices and strikes use index points. Let $r(t)$ and $q(t)$ be deterministic pricing rate and yield, with

$$D(t) = \exp\left[-\int_0^t r(u)du\right], \qquad F(t) = S_0 \exp\left[\int_0^t (r(u) - q(u))du\right].$$

$W^{\mathbb{Q}}$ denotes risk-neutral Brownian motion; Φ and ϕ are the standard normal distribution and density. Volatility is annualized in decimal units; figures use percent. Local variance has units of inverse year.

4.1 Local versus implied volatility

The model dynamics are

$$\frac{dS_t}{S_t} = (r(t) - q(t))dt + \sigma_{\mathrm{LV}}(S_t, t)dW_t^{\mathbb{Q}}.$$

The coefficient is a deterministic function of level and time, while the index path remains stochastic. A constant coefficient recovers the Black–Scholes special case. Level and maturity dependence allow the calibrated model to encode a smile and term structure.

Implied volatility instead reproduces one European price under Black's formula:

$$C(K,T) = D(T)\{F(T)\Phi(d_1) - K\Phi(d_2)\},$$

$$d_1 = \frac{\log(F(T)/K) + \sigma_{\text{imp}}^2 T/2}{\sigma_{\text{imp}}\sqrt{T}}, \qquad d_2 = d_1 - \sigma_{\text{imp}}\sqrt{T}.$$

Put prices follow parity. The implementation uses stable time-value formulas and preserves inversion failure statuses at noninvertible midpoints. It does not manufacture a positive IV to make every quote appear supported. Total implied variance and log forward-moneyness are

$$y = \log(K/F(T)), \qquad w(y,T) = \sigma_{\text{imp}}(y,T)^2 T.$$

An implied volatility is a constant-volatility price representation; a local volatility specifies instantaneous diffusion. They are not interchangeable.

4.2 Forward Dupire equation and density

For a sufficiently regular European call surface, the deterministic-carry forward equation is

$$C_T = \tfrac{1}{2}\sigma_{\text{LV}}^2(K,T)K^2 C_{KK} - (r(T) - q(T))KC_K - q(T)C.$$

Hence

$$\sigma_{\text{LV}}^2(K,T) = \frac{2\{C_T + q(T)C + (r(T) - q(T))KC_K\}}{K^2 C_{KK}}.$$

All derivatives refer to options observed on the same valuation date. C_T moves along that date's maturity structure at fixed strike; it is not a difference between prices on two historical quote dates. Under the regular risk-neutral setting, $C_{KK} = D(T)f_{S_T}(K)$ is discounted terminal density. Small curvature makes the ratio sensitive to interpolation and price errors. Admissibility and derivative support therefore precede interpretation of a local-volatility graph. This is the forward-price construction associated with Dupire [1].

Strike curvature of a cross-section and backward spot gamma are distinct: C_{KK} differentiates prices across contracts, whereas V_{SS} differentiates one fixed contract with respect to its initial state.

4.3 Carry normalization and its derivative convention

Set

$$z = K/F(T), \qquad c(z,T) = C(F(T)z,T)/(D(T)F(T)).$$

At fixed physical strike, differentiating this representation gives

$$C_T = DF\{c_T - (r-q)zc_z - qc\}, \quad C_K = Dc_z, \quad C_{KK} = (D/F)c_{zz}.$$

Substitution cancels deterministic carry and yields

$$c_T = \tfrac{1}{2}a(z,T)z^2c_{zz}, \qquad a(z,T) = \frac{2c_T}{z^2c_{zz}}, \qquad c(z,0) = (1-z)^+.$$

Here the normalized derivative c_T holds z fixed, and $a(z,T) = \sigma^2_{\mathrm{LV}}(F(T)z,T)$. It describes a future state under one valuation date's surface. Write $a_y(y,T) = a(e^y,T)$ when using log coordinates.

5 Daily surface calibration

5.1 Put–call parity and spread weighting

For each date and expiry,

$$C(K,T) - P(K,T) = D(T)\{F(T) - K\}.$$

The primary forward fit uses positive two-sided call–put pairs inside $|\log(K/S)| \le 0.03$ and needs at least six selected pairs. Other parity windows and rate cases are diagnostics; the frozen primary pricing rate is 5%. A forward can be fitted even when original parity bands are mutually incompatible; those statuses remain recorded rather than being hidden.

An out-of-the-money side is selected at each strike and converted to an equivalent call. For a put, adding the deterministic parity offset shifts bid, ask and midpoint equally, preserving its original half-spread $h_j = (C_{\mathrm{ask},j} - C_{\mathrm{bid},j})/2$. Noninvertible IV midpoints can still be price-space calibration observations. Their status remains visible.

5.2 AH implicit construction and objective

AH fits each expiry sequentially. Let p_i denote log-volatility controls and $\nu_i(z) = \exp[W_i(z)p_i]$ their interpolated proxy volatility. On the finite normalized-strike grid, the implicit expiry step is

$$M_ic_i = c_{i-1}, \quad M_i = I - \Delta T_iA_i, \quad A_i = \tfrac{1}{2}\nu_i(z)^2z^2\partial_{zz}.$$

The calibrated objective is

$$\min_{p_i} \sum_j \left[\frac{D_iF_ic_i(K_j/F_i;p_i) - C_{\mathrm{mid},j}}{h_j}\right]^2 + \lambda\|Lp_i\|_2^2.$$

L is the first-difference matrix of adjacent log controls, not a second spatial derivative. The frozen value is $\lambda = 1$. Residuals are reported in original half-spread units and index points, alongside expiry fit counts, conditioning and original-band exclusions. The objective fits quoted prices; it does not optimize historical hedge Gain.

The normalized grid is uniform in log strike with 8,000 intervals and half-width 1.0. Each expiry has at most 31 controls spanning its quoted log strikes. Proxy volatility is bounded between 0.005 and 3.0; the solver allows 300 evaluations and checks optimizer success. These are bounds on proxy parameters, not caps on recovered local volatility.

5.3 Proxy parameters and recovered local variance

Inside an expiry interval, the price construction is an implicit resolvent: $c(\tau) = (I - \tau A_i)^{-1} c_{i-1}$. Differentiation gives

$$\partial_\tau c = (I - \tau A_i)^{-1} A_i c(\tau).$$

Consequently the calibrated proxy ν_i^2 need not equal the recovered ratio $2c_T/(z^2 c_{zz})$. The coefficient implementation uses native density and time derivatives, including logarithmic positive calculations for small wing quantities. It does not estimate curvature by twice differentiating a piecewise-linear price interpolant. The implicit construction follows the AH approach [2], with the specified regularization and diagnostic conventions.

Large recovered coefficients remain visible. Changing a plotting domain is an explicitly declared display choice; it is not an arbitrage repair or a cap on the model. Maturity-side derivatives and coefficient jumps are retained at expiry boundaries.

5.4 Short-end diffusion blend

Before the first fitted maturity T_1, the hedge diffusion uses

$$\tilde{a}_y(y,t) = (1-\omega(t))a_y(y,t) + \omega(t)(G_\rho * a_y)(y,t), \qquad \omega(t) = (1-t/T_1)^2, \quad 0 < t < T_1.$$

The Gaussian filter acts on variance at native log-state nodes, with radius $\rho = 0.0005$ and finite truncation at four radii. From T_1 onward the raw coefficient is retained. Only positive coefficient times are supported. This blend changes the diffusion relative to the native calibrated price surface. Its price and Greek effects require separate checks.

5.5 Quote-PCHIP and constrained SSVI diagnostics

The independent quote construction interpolates total variance with PCHIP across y and linearly across expiries. Invalid IV observations split contiguous strike support. It does not bridge gaps or extrapolate a short end from unsupported quotes. The smooth total-variance Dupire representation is

$$a_y = \frac{w_T}{g}, \qquad g = \left(1 - \frac{y w_y}{2w}\right)^2 - \frac{w_y^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w_{yy}}{2}.$$

Availability requires supported derivatives, positive total variance, nonnegative calendar derivative, a sufficiently positive density denominator, and stable derivative stencils. Unsupported or unstable local-volatility points remain gaps. The construction is diagnostic and does not replace the historical hedge diffusion.

The saved constrained SSVI comparison [3] uses

$$w(y,\theta) = \frac{\theta}{2}\left[1 + \rho_{\mathrm{SVI}}\varphi(\theta)y + \sqrt{(\varphi(\theta)y + \rho_{\mathrm{SVI}})^2 + 1 - \rho_{\mathrm{SVI}}^2}\right],$$

$$\varphi(\theta) = \frac{\eta}{\sqrt{\theta(1+\theta)}}.$$

ATM variance starts from zero, is nondecreasing and is interpolated linearly in time. The constructor requires $|\rho_{\mathrm{SVI}}| < 1$, $\eta \geq 0$ and $\eta^2(1 + |\rho_{\mathrm{SVI}}|) < 4$. These give the implemented sufficient

butterfly restrictions, while the chosen shape and monotone ATM variance satisfy the calendar condition. Maturity derivatives are one-sided at pillars; evaluation stops at the final maturity. The restricted shared-shape family fits the inspected quotes less closely than AH and was not used as a frozen hedge method. Browsing loads a saved SSVI fit only on its recorded date.

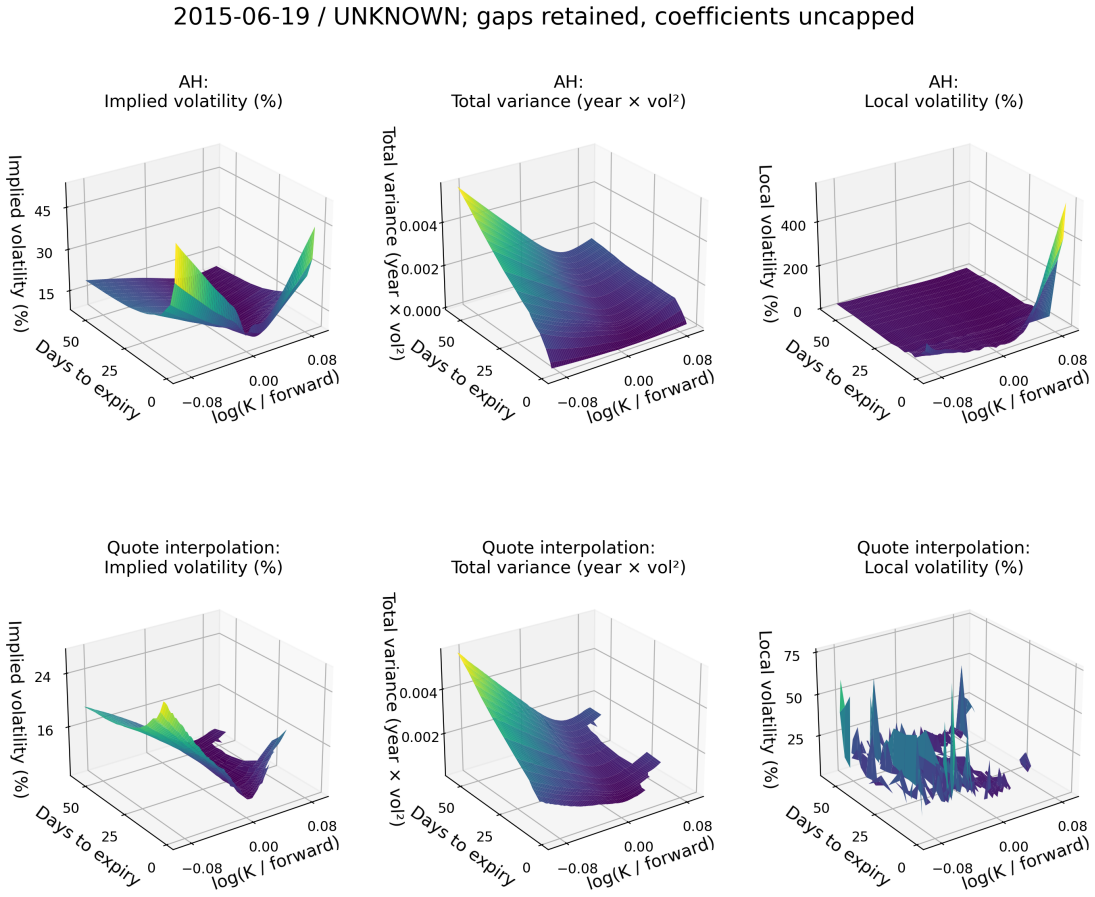


Figure 1: Daily saved AH and quote-based surface comparison. June 19, 2015; root UNKNOWN; log forward-moneyness domain −0.09 to +0.09. Volatilities are annualized percent; total variance is sigma squared times years. Coefficients are uncapped and quote gaps remain visible.

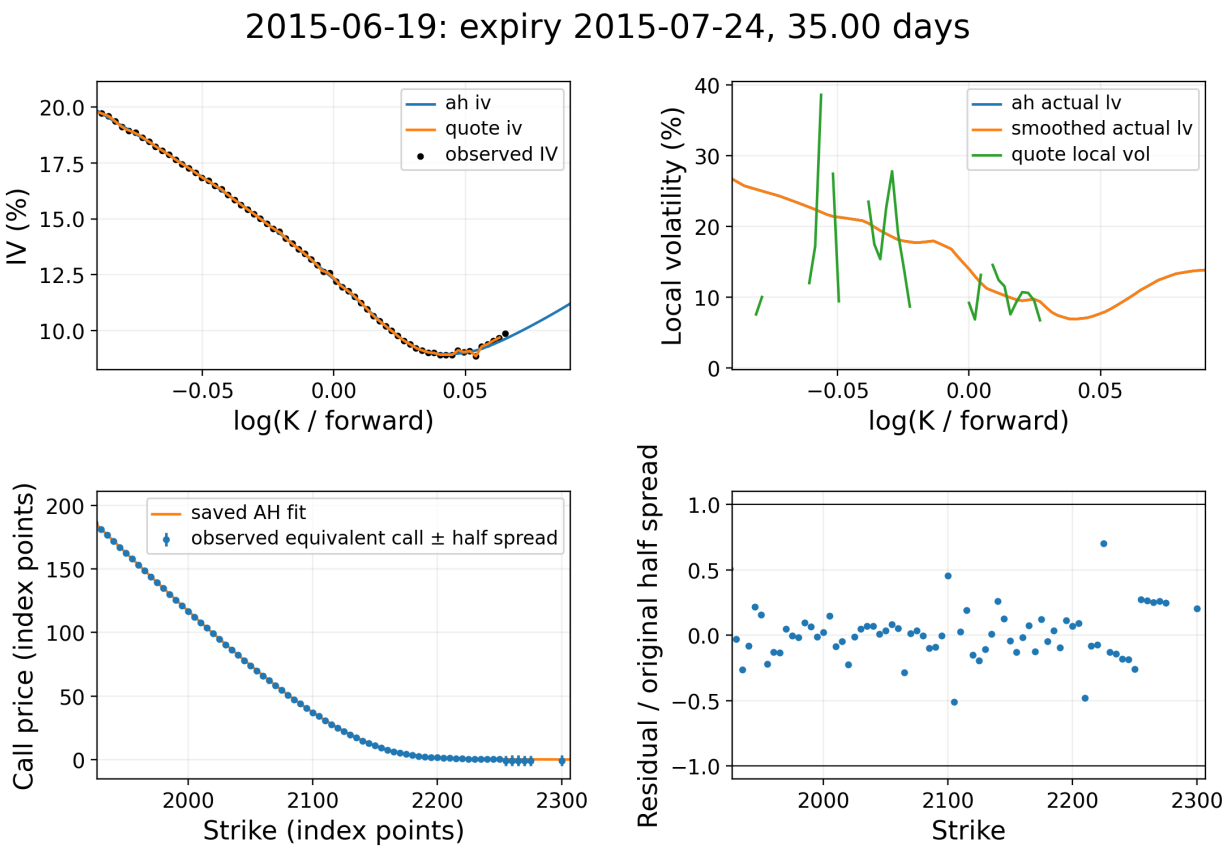


Figure 2: Same-date July 24, 2015 maturity slice (35 days), equivalent-call quote fit and residuals in original half-spread units. The display uses the same declared log-strike domain.

6 Backward pricing and spot Greeks

For a normalized state x, define $\tilde{a}(x,t) = \tilde{a}_y(\log x, t)$.

For a fixed European contract the physical-state backward equation is

$$V_t + (r(t) - q(t))SV_S + \tfrac{1}{2}\sigma_{\mathrm{LV}}^2(S,t)S^2V_{SS} - r(t)V = 0, \qquad V(S,T) = (S-K)^+$$

for a call; a put has $(K-S)^+$ terminal payoff. The hedge solver replaces $\sigma_{\mathrm{LV}}^2(S,t)$ by the blended coefficient $\tilde{a}(S/F(t),t)$. To remove deterministic carry, set $X_t = S_t/F(t)$ and $Y_t = \log X_t$. Itô's formula gives

$$dX_t = X_t\sqrt{\tilde{a}(X_t,t)}dW_t^{\mathbb{Q}}, \qquad dY_t = -\tfrac{1}{2}\tilde{a}_y(Y_t,t)dt + \sqrt{\tilde{a}_y(Y_t,t)}dW_t^{\mathbb{Q}}.$$

For normalized conditional payoff U, the generator is

$$U_t + \tfrac{1}{2}\tilde{a}_y(y,t)(U_{yy} - U_y) = 0, \qquad U(y,T) = (e^y - k)^+, \quad k = K/F(T).$$

With $A(t,T) = D(T)F(T)/D(t)$, the price and Greeks are

$$V(S,t) = A(t,T)U(\log(S/F(t)),t), \quad \Delta = \frac{AU_y}{S}, \quad \Gamma = \frac{A(U_{yy}-U_y)}{S^2}.$$

The code interpolates solved values by a cubic spline in log state and takes its first and second derivatives. Spot Greeks hold the coefficient function and deterministic carry fixed. They do not recalibrate after a spot bump. A chronological rebalance uses a different current-date model, which is a separate operation.

6.1 Discretization and numerical error controls

The frozen hedge solver has 24,000 log-state intervals, domain half-width 0.75 and 128 nominal steps per day. It uses centered spatial differences and midpoint coefficient evaluations. Most steps are Crank–Nicolson; the first two nominal steps backward from the payoff are split into implicit Rannacher half-steps. The time mesh includes maturity breaks and is graded with power three toward zero before the first pillar. Actual step counts are recorded.

At the finite state endpoints the normalized terminal-payoff boundary values are held fixed. The PDE domain stays within the native coefficient domain; unsupported coefficient extrapolation and zero-time requests fail clearly. Space, time, domain and state shifts are refined separately. A smaller price error on one grid does not independently certify gamma convergence.

7 Hedge specifications

7.1 Observed-IV Black benchmark

At current spot S, forward F, discount D and maturity T, Black spot delta and vega are

$$\Delta_B = \frac{DF}{S}\{\Phi(d_1) - \mathbf{1}_{\{\text{put}\}}\}, \qquad \mathcal{V}_B = DF\phi(d_1)\sqrt{T}.$$

The Black spot derivative keeps F/S and IV fixed. Vega is per unit decimal annual volatility, so multiplying by 0.01 gives sensitivity per volatility percentage point.

7.2 Smile movement conventions

Let σ_y be the slope of fitted implied volatility with respect to $\log(K/F)$. The hedge rules retain their own availability checks:

Method	Definition
AH PDE	Saved backward-PDE spot derivative under the blended local variance
Sticky strike	Black delta at fitted IV, with fitted IV held fixed at strike
Sticky delta	Fitted-IV Black delta minus fitted-IV vega times σ_y/S
LV smile	Observed-IV Black delta plus observed-IV vega times σ_y/K
Empirical MV	Observed-IV Black delta plus a past-data fitted correction
Unhedged	Zero index position

Sticky delta uses an unadjusted normalized forward-call-delta coordinate. The chain rule and a finite spot bump are compared, and the coordinate must be invertible. The LV-smile correction uses the fitted smile slope but observed-IV vega; it is an approximation, not an AH PDE delta. Native-cell stencil widths and their coarse/fine comparisons remain explicit. Disagreement or insufficient observed support is an exclusion rather than a fallback hedge.

7.3 Past-only empirical correction

The unconstrained one-period minimum-variance motivation is

$$h^* = \frac{\mathrm{Cov}(\Delta C, \Delta S)}{\mathrm{Var}(\Delta S)}.$$

The implemented Hull–White-style feature model [4] uses

$$x_i = \frac{\mathcal{V}_{B,i}}{S_i\sqrt{T_i}}\begin{bmatrix}1 & \Delta_{B,i} & \Delta_{B,i}^2\end{bmatrix}, \qquad \Delta_{\mathrm{MV},i} = \Delta_{B,i} + x_i\hat{\beta}.$$

Ordinary least squares fits $\Delta C_i - \Delta_{B,i}\Delta S_i$ against $x_i\Delta S_i$, with no additional intercept, separately by root and option kind. The rolling window contains up to 60 past entry dates and requires at least ten dates and 60 rows, full feature rank and acceptable conditioning. The condition-number limit is 10^8. Every admitted outcome endpoint must be strictly earlier than the prediction timestamp. Incomplete, unavailable or warm-up fits do not silently substitute another hedge.

This objective uses raw observed price changes. Funding and execution costs are applied afterward to the multi-session hedge ledger. It is an empirical specification rather than an exact conditional-variance guarantee for every holding period or funded account.

8 Chronological paths and self-financing accounting

Entry selection uses current-date observations and deduplicates repeated target buckets that choose the same contract. Future mark or strategy availability is not used to screen an entry. Holding lengths count reference sessions. Missing observations, early-close exclusions and strategy failures interrupt paths with recorded dates and reasons.

For the unit short option, the target index position is the selected delta. Let O_i be the option position (-1 while held, zero after exit), H_i the index position, B_i cash and M_i the observed option mark. Over an interval,

$$B_i^- = B_{i-1}^+\exp(r_f\Delta t_i), \qquad B_i^+ = B_i^- - \Delta O_i M_i^{\mathrm{fill}} - \Delta H_i S_i^{\mathrm{fill}} - \mathrm{fees}_i.$$

The account begins with zero initial capital. Entry option proceeds fund the hedge purchase; any remainder is cash, possibly negative. Funding uses the opening cash sign, although the declared lending and borrowing rates are both 5%. At exit both the option and index hedge are liquidated. Equity marked to observed prices is $E_i = B_i + O_iM_i + H_iS_i$.

The midpoint case uses midpoint option fills. The option-spread case sells at bid and buys back at ask. The final case adds 1 bp of absolute traded hedge notional, including entry, rebalancing and liquidation. The synthetic index is filled at the observed mark in every case, without an invented executable index spread. Dividend cash is zero and is not inferred from parity yield.

Accounting separates option-mark P&L, index P&L, interest and costs. For the unit conventions,

$$\Delta E_i = O_{i-1}\Delta M_i + H_{i-1}\Delta S_i + \mathrm{interest}_i - \mathrm{costs}_i.$$

The ledger checks both the incremental and cumulative identities. In the completed study, the largest stored reconciliation is 4.60×10^{-12} index points. This is an accounting residual, not a pricing or hedge-performance error. Vectorized path outcomes are cross-checked against the scalar ledger on retained representative paths.

9 Statistical comparison and uncertainty

Each candidate is paired with a completed Black outcome on the same entry, schedule and execution case. The common-entry sample is the intersection shared by every method, including the unhedged control. A ready path for one method can fail to enter a pair if its Black counterpart is unavailable. Coverage retains all selected entries and distinguishes these populations.

For funded P&L P_i around zero,

$$\text{Gain} = 1 - \frac{\sum_i P_{i,\text{candidate}}^2}{\sum_i P_{i,\text{Black}}^2}.$$

A negative Gain means more squared funded error. It is neither an investment return nor a reduction of demeaned variance. Squaring around zero includes mean bias and gives substantial weight to tail outcomes. RMS, mean error, MAE and paired absolute-error comparisons provide additional diagnostics. Gain is unavailable when the matched Black SSE is effectively zero.

A stationary date-block bootstrap keeps all contracts from an entry date together and uses 500 replicates, seed 20261008, and mean block lengths 10, 20 and 40 sessions. Blocks are circular and address dependence from overlapping holding periods. The percentile interval uses the saved 2.5% and 97.5% quantiles. Models, entry selection and empirical coefficients are not fitted again within a replicate, so the intervals are conditional on the retained historical pipeline. They are not a guarantee for a future execution regime.

Grouping by year, maturity, moneyness and Black-delta bucket uses saved outputs. An arbitrary entry-date range sums daily candidate SSE, matched Black SSE and counts before computing descriptive Gain. It never averages daily percentages. No full-history interval is attached to such a range, and common-entry daily sufficient statistics were not saved, so common-entry range estimates are not manufactured.

10 Numerical verification

Analytical Black and square-root CEV controls check pricing, Greeks, payoff parity, transitions and funded accounting. The retained suite is data-free: it generates small explicitly artificial quote histories and calls the same core objects. Market evidence remains a separate producer-pinned scope.

On June 19, 2015, 30 contracts, four blend radii, six PDE configurations and three initial spots give 2,160 sampled states. No sampled evaluation failures, price/delta-bound flags or negative-gamma flags were recorded. At the original spot and historical blend radius, the largest 24k-to-48k delta change was 3.36×10^{-6}. Gamma was more sensitive to grid position, blend radius and bump width. Selected calls on March 16, 2020 and November 9, 2023 reproduced production deltas within 7.91×10^{-13} on their original grids.

The June calibration-vega panel uses symmetric parallel quote-IV shocks of 1, 2 and 5 basis points, holding carry fixed. It covers 18 contracts and seven calibrations. Of 597 quotes, 566 support the shock; 31 noninvertible midpoints remain unchanged and participate in the price fit. The largest native/PDE relative vega difference is 0.993% for this declared direction. It is not a validation of every independent calibration pillar.

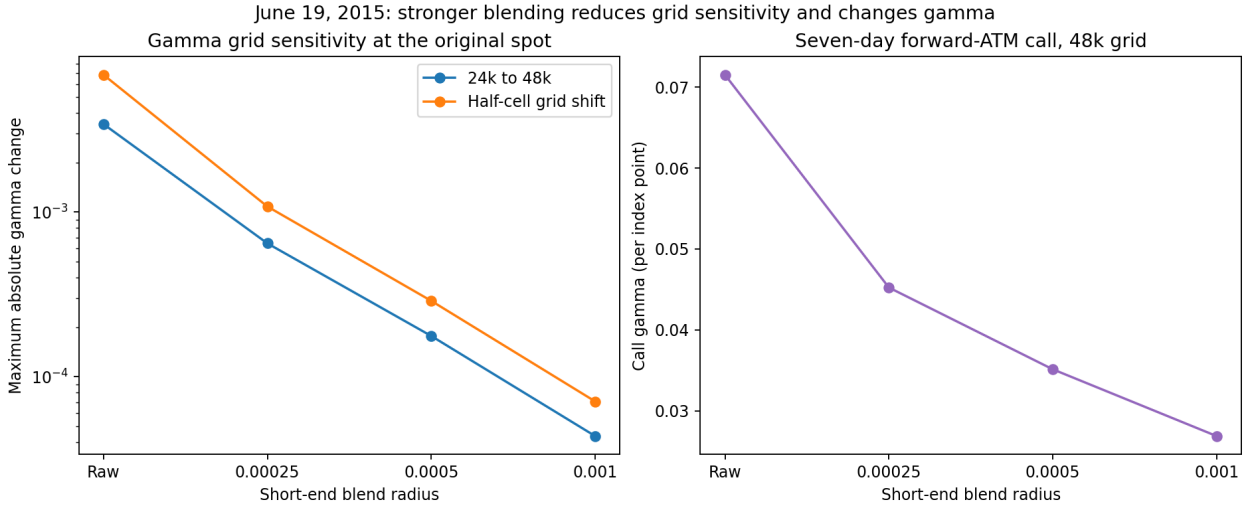


Figure 3: Selected June Greek and blend sensitivity, from retained numerical outputs.

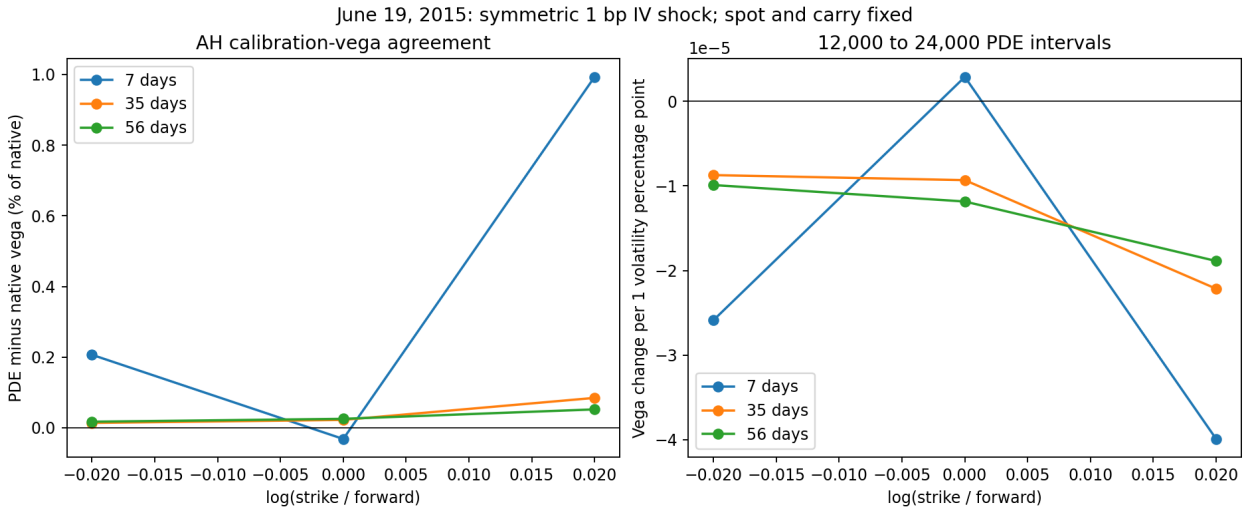


Figure 4: Selected parallel quote-IV calibration-vega comparison, with unchanged carry.

Seven-date smile replays cover 928 decisions with zero status mismatches. Fourteen empirical membership checks retain endpoints strictly before the prediction timestamp. These panels support their stated scopes; they do not independently validate every Greek in the eleven-year history.

11 Full-history findings

The completed study contains 44,500 entries, 2,768 reference dates, 3,879,048 compared strategy/schedule/scenario outcomes and zero pending paths. Outcomes reuse entries and must not be counted as independent observations. The five-session, daily-rebalanced midpoint comparison is:

Method	Paired entries	Pair Gain (%)	Common Gain (%)	Common 95% CI (%)
AH PDE	39,252	-25.58	-30.16	[-55.50, -7.57]
Empirical MV	40,230	-8.40	-7.92	[-25.78, +9.08]
LV smile	33,887	-12.38	-12.40	[-30.03, +4.53]
Sticky delta	33,887	-136.71	-136.74	[-188.49, -91.90]
Sticky strike	33,887	+0.15	+0.15	[-0.04, +0.37]

The common-entry sample has 33,763 entries across 2,406 entry dates. Its AH Gain of −30.16% implies a candidate/Black SSE ratio of 1.3016. Empirical MV and LV smile have intervals spanning zero. Sticky delta has substantially larger squared errors, and sticky strike's small positive estimate also has an interval spanning zero. The conclusion is adverse for AH under these conventions. It does not establish a general hedge-performance improvement.

All fifteen AH schedule/execution point estimates are negative. Some common-entry intervals include zero; others remain adverse. For midpoint execution, common-entry AH Gain is −18.42%, −30.16%, −13.71%, −26.10% and −13.82% for 1/1, 5/1, 5/5, 10/1 and 10/5 respectively. The appendix gives all five candidate methods, schedules and execution scenarios, with counts and saved 20-session common-entry intervals. The accompanying compact CSV also retains pairwise intervals, all three block lengths and the unhedged control.

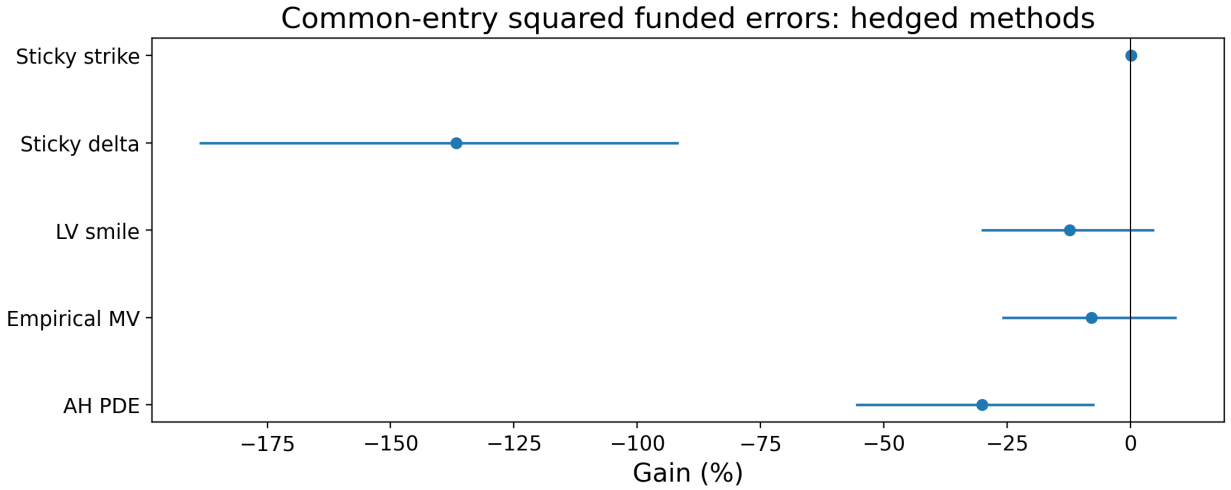


Figure 5: Common-entry funded Gain and saved 95% intervals: five-session holding, daily rebalance, midpoint execution and mean bootstrap block 20 sessions. The figure shows hedged methods; the tables retain the unhedged control.

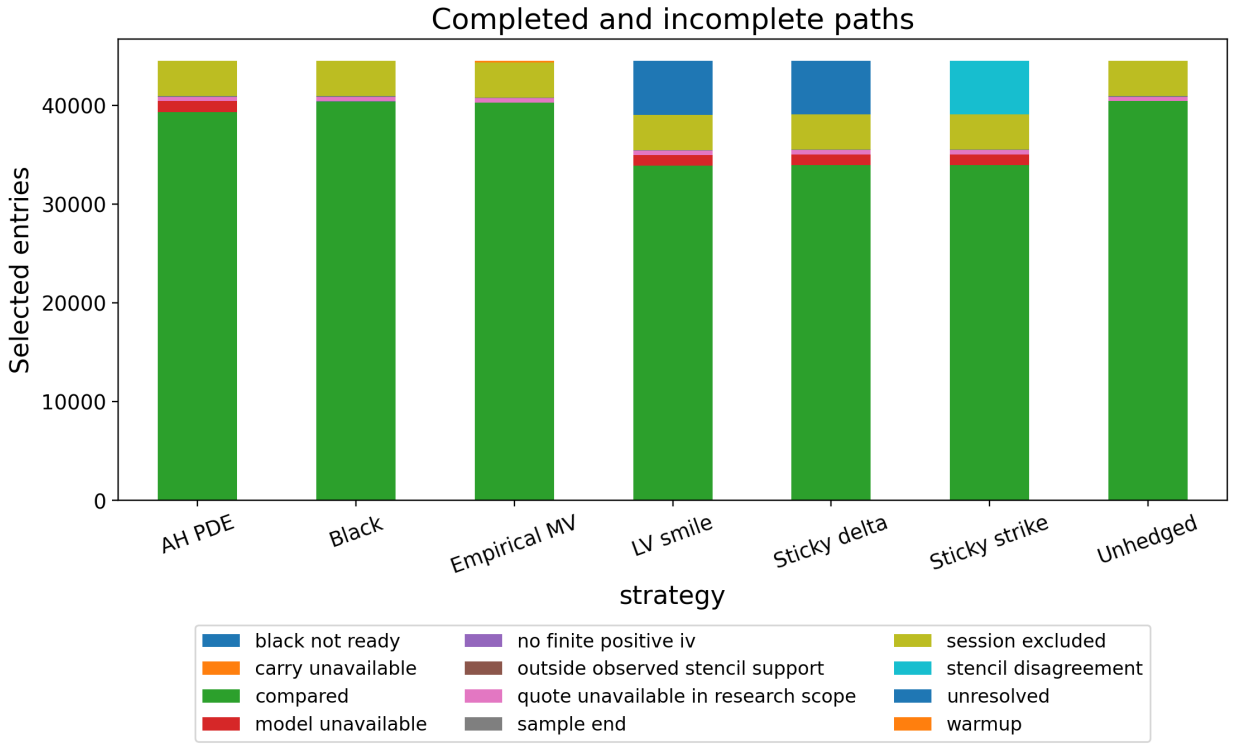


Figure 6: Completed and incomplete coverage for the same five-session daily-rebalanced midpoint scope. Every method retains all 44,500 selected entries in its status counts.

A tight fit to today's European prices restricts the current risk-neutral marginals. It does not establish how tomorrow's observed smile moves, nor whether the chosen spot hedge minimizes realized multi-session funded errors. The distinction is consistent with the mixed or adverse historical comparison. The evidence does not identify a single cause of full-history underperformance.

12 Carry exclusions and selected tail diagnostics

Thirty dates have no AH fit because carry is unavailable for at least one expiry: 20 dates in 2016 and ten in 2022. There are 31 bad expiry records, including two on May 27, 2022. Each records insufficient selected parity pairs (three to five), an unavailable conditional fit and unavailable carry. These are pre-optimization support failures; no expiry is silently removed to make a daily fit succeed. Available pairs outside the selected parity window do not satisfy its support requirement.

For six declared dates—March 2, March 3, March 16 and March 26, 2020; September 19 and November 9, 2023—the diagnostic selects each AH/sticky-delta contract with the largest signed midpoint excess SSE over matched Black. Twelve candidate/method paths are replayed against Black in all three cases: 36 paired scenario checks, or 72 scalar ledgers. The maximum discrepancy in P&L, costs, funding, turnover and component P&L is 1.49×10^{-12} index points. This is an outcome-selected diagnostic, not new performance data.

The selected contract remains the midpoint-selected contract in each scenario. Daily candidate SSE, Black SSE and pair counts on a scenario row are computed from that row's execution case. Thus spread/fee rows do not repeat midpoint statistics under an unlabeled generic heading. For March 16, 2020 AH, the 18 midpoint pairs have candidate/Black daily SSE 153,541.47/93,737.10; option-spread pairs have 135,005.20/79,435.75; and spread-plus-fee pairs have 134,048.80/78,731.80. The counts coincide here, while funded SSE differs.

For a selected pair, candidate minus Black P&L separates into first-interval and later-interval

hedge contributions, funding difference and cost difference. Observed option-mark P&L is the same within the pair and cancels. Replayed later contributions use actual saved rebalances. Their signs and magnitudes vary by date; entry-date Greek refinements alone do not explain entire paths. Full contract identities, exclusions, intervals and scenario statistics remain in the curated and producer-pinned diagnostic tables.

13 Interpretation and limitations

The project delivers a daily calibrated surface, recovered local variance, backward pricing and Greek objects, transparent hedge conventions and a chronological historical comparison. The principal finding is that AH did not provide a stable hedge advantage over Black under the declared assumptions. The result is retained even though the daily price fit is close.

The available data do not certify roots, snapshot provenance or official fixings. Pricing and funding rates are assumed constants. Synthetic fractional index execution, zero hedge-account dividend cash and provisional settlement identity limit interpretation of the funded ledger. Exclusions alter available samples, and common-entry matching cannot remove all such selection effects.

The history is retrospective; the bootstrap is conditional on frozen estimates and does not re-estimate the model. Selected numerical refinements do not certify every historical delta or gamma. Broader gamma validation, all-history Greek verification, more flexible SSVI calibration and deployable hedge execution are outside the completed evidence. No model tuning was performed to recover an unsupported performance target.

14 Reproducibility and artifact identity

The frozen fingerprint is:

`09c095fb8a89f2ab811b2bddb393ce0a9258a8d013095b8c23a9960dd6b4a232`

The SHA-256 of `freeze.json` is:

`b54c0bb421abb14fa64e1e223d796982312bbc34a7c02bb182829538a6207c7e`

These identify the completed original study. Its recorded environment is Python 3.11.15, NumPy 2.4.3, pandas 2.3.3 and SciPy 1.17.1. Producer audits pin consumed model, quote, carry, decision and summary files.

The public source is available at github.com/KVInator/local-vol-spx. Private raw data and the multi-year cache are not distributed. The public repository includes editable report source, this PDF, curated figures, summary CSV files and their provenance. The data-free unit suite and report rendering can run without the private study. Recreating empirical figures or browsing daily cached surfaces requires the recorded study and configured paths.

Use `PYTHONPATH=src python -m unittest discover -s tests` from the root. The notebooks share `config/notebooks.json`. An unavailable requested date remains unavailable through display-control changes; explicitly choosing a saved date restores synchronized content. Missing optional numerical/SSVI folders are reported without blocking the required historical study. Present malformed audits, mismatched study identities and changed producer outputs remain errors.

The PDF is rendered from this authoritative Markdown by the single documented Pandoc/Tectonic path. Temporary TeX files, logs, caches and inspection images are confined to `scratch/`. Figure provenance records the selected date, domain, maturity, units, schedule, sample and input/output hashes.

15 References

1. Bruno Dupire (1994), *Pricing with a Smile*, Risk 7(1), 18–20. Publisher reprint.
2. Jesper Andreasen and Brian Huge (2011), *Volatility Interpolation*, Risk. Publisher page.
3. Jim Gatheral and Antoine Jacquier (2014), *Arbitrage-free SVI volatility surfaces*, Quantitative Finance 14(1), 59–71; author manuscript arXiv:1204.0646. Author manuscript.
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16 Appendix: saved schedule and execution comparisons

Schedules are holding/rebalance reference sessions. Gain and interval bounds are percentages. Pair N is the candidate–Black paired entry count. Common Gain and its 95% interval use the intersection shared by all methods for the same schedule and execution case, with 500 replicates and mean block 20 sessions. These are saved estimates, not a new bootstrap run.

Common-entry counts are the same across the three execution cases:

Schedule	Common entries
1/1	41,771
5/1	33,763
5/5	38,643
10/1	24,882
10/5	32,365

16.1 Midpoint execution

Schedule	Method	Pair N	Pair Gain (%)	Common Gain (%)	Common 95% CI (%)
1/1	AH PDE	43,185	-18.66	-18.42	[-44.35, +2.43]
1/1	Empirical MV	43,527	+4.64	+4.93	[-8.89, +16.71]
1/1	LV smile	41,903	-6.93	-6.93	[-31.00, +11.99]
1/1	Sticky delta	41,903	-136.35	-136.37	[-166.89, -109.78]
1/1	Sticky strike	41,903	+0.05	+0.05	[-0.01, +0.12]
5/1	AH PDE	39,252	-25.58	-30.16	[-55.50, -7.57]
5/1	Empirical MV	40,230	-8.40	-7.92	[-25.78, +9.08]
5/1	LV smile	33,887	-12.38	-12.40	[-30.03, +4.53]
5/1	Sticky delta	33,887	-136.71	-136.74	[-188.49, -91.90]
5/1	Sticky strike	33,887	+0.15	+0.15	[-0.04, +0.37]
5/5	AH PDE	39,957	-12.99	-13.71	[-33.09, +4.33]
5/5	Empirical MV	40,299	+0.17	+0.28	[-9.68, +9.55]
5/5	LV smile	38,775	-5.87	-5.88	[-22.11, +9.96]
5/5	Sticky delta	38,775	-73.11	-73.11	[-89.24, -56.60]
5/5	Sticky strike	38,775	+0.02	+0.02	[-0.01, +0.04]
10/1	AH PDE	33,434	-23.53	-26.10	[-56.12, -0.01]
10/1	Empirical MV	34,939	-11.28	-10.96	[-39.21, +10.23]
10/1	LV smile	24,980	-2.23	-2.25	[-24.08, +17.28]
10/1	Sticky delta	24,980	-157.74	-157.79	[-194.61, -126.91]
10/1	Sticky strike	24,980	+0.04	+0.04	[-0.39, +0.43]
10/5	AH PDE	34,662	-12.59	-13.82	[-31.64, +3.94]
10/5	Empirical MV	35,362	-3.17	-3.34	[-11.58, +5.96]
10/5	LV smile	32,483	-4.74	-4.74	[-18.09, +9.14]
10/5	Sticky delta	32,483	-69.97	-69.97	[-85.20, -57.71]
10/5	Sticky strike	32,483	-0.08	-0.08	[-0.18, +0.02]

16.2 Option bid–ask execution

Schedule	Method	Pair N	Pair Gain (%)	Common Gain (%)	Common 95% CI (%)
1/1	AH PDE	43,185	-12.87	-12.52	[-34.72, +4.99]
1/1	Empirical MV	43,527	+3.98	+4.19	[-8.18, +14.02]
1/1	LV smile	41,903	-3.79	-3.79	[-24.40, +12.02]
1/1	Sticky delta	41,903	-108.34	-108.35	[-128.33, -87.50]
1/1	Sticky strike	41,903	+0.04	+0.04	[+0.01, +0.07]
5/1	AH PDE	39,252	-23.52	-28.44	[-56.23, -5.22]
5/1	Empirical MV	40,230	-8.60	-8.26	[-26.76, +8.41]
5/1	LV smile	33,887	-12.50	-12.51	[-31.42, +5.04]
5/1	Sticky delta	33,887	-125.99	-126.00	[-171.95, -84.65]
5/1	Sticky strike	33,887	+0.14	+0.14	[-0.06, +0.38]
5/5	AH PDE	39,957	-13.00	-13.74	[-34.23, +5.17]
5/5	Empirical MV	40,299	-0.56	-0.48	[-11.16, +8.96]
5/5	LV smile	38,775	-6.22	-6.22	[-23.74, +10.24]
5/5	Sticky delta	38,775	-70.36	-70.37	[-86.09, -54.59]
5/5	Sticky strike	38,775	+0.03	+0.03	[-0.00, +0.06]
10/1	AH PDE	33,434	-26.34	-30.45	[-65.56, -2.16]
10/1	Empirical MV	34,939	-13.85	-14.46	[-42.92, +6.86]
10/1	LV smile	24,980	-6.56	-6.58	[-31.88, +14.47]
10/1	Sticky delta	24,980	-149.08	-149.11	[-185.23, -118.06]
10/1	Sticky strike	24,980	-0.00	-0.00	[-0.46, +0.40]
10/5	AH PDE	34,662	-13.95	-15.31	[-33.87, +3.92]
10/5	Empirical MV	35,362	-4.57	-4.84	[-13.65, +5.05]
10/5	LV smile	32,483	-6.21	-6.21	[-20.85, +8.74]
10/5	Sticky delta	32,483	-67.69	-67.69	[-83.67, -55.22]
10/5	Sticky strike	32,483	-0.07	-0.07	[-0.17, +0.04]

16.3 Option bid–ask plus hedge fee

Schedule	Method	Pair N	Pair Gain (%)	Common Gain (%)	Common 95% CI (%)
1/1	AH PDE	43,185	-12.30	-11.95	[-33.19, +4.92]
1/1	Empirical MV	43,527	+3.75	+3.95	[-7.89, +13.54]
1/1	LV smile	41,903	-3.66	-3.66	[-23.43, +11.44]
1/1	Sticky delta	41,903	-103.41	-103.43	[-122.03, -83.27]
1/1	Sticky strike	41,903	+0.03	+0.03	[+0.00, +0.06]
5/1	AH PDE	39,252	-23.27	-28.24	[-56.24, -5.03]
5/1	Empirical MV	40,230	-8.78	-8.46	[-27.24, +8.14]
5/1	LV smile	33,887	-12.72	-12.73	[-31.74, +4.59]
5/1	Sticky delta	33,887	-123.42	-123.43	[-167.94, -83.03]
5/1	Sticky strike	33,887	+0.14	+0.14	[-0.07, +0.37]
5/5	AH PDE	39,957	-13.13	-13.86	[-34.35, +5.35]
5/5	Empirical MV	40,299	-0.70	-0.62	[-11.39, +8.84]
5/5	LV smile	38,775	-6.38	-6.38	[-24.14, +10.22]
5/5	Sticky delta	38,775	-69.90	-69.91	[-85.67, -54.11]
5/5	Sticky strike	38,775	+0.03	+0.03	[-0.01, +0.06]
10/1	AH PDE	33,434	-26.73	-31.18	[-67.75, -2.69]
10/1	Empirical MV	34,939	-14.37	-15.09	[-43.63, +6.09]
10/1	LV smile	24,980	-7.59	-7.60	[-33.98, +13.83]
10/1	Sticky delta	24,980	-146.59	-146.62	[-182.40, -115.97]
10/1	Sticky strike	24,980	-0.02	-0.02	[-0.48, +0.38]
10/5	AH PDE	34,662	-14.19	-15.57	[-34.47, +3.90]
10/5	Empirical MV	35,362	-4.81	-5.09	[-14.08, +4.86]
10/5	LV smile	32,483	-6.50	-6.50	[-21.27, +8.59]
10/5	Sticky delta	32,483	-67.29	-67.30	[-83.37, -54.57]
10/5	Sticky strike	32,483	-0.07	-0.07	[-0.17, +0.03]